Probability

Section 1: Discrete Random Variables & Variance

1 A discrete random variable X has distribution: $P(X=0)=0.1, P(X=1)=0.4, P(X=2)=0.5$. What is $E(X)$?

a. 1.4

b. 1.0

c. 1.5

d. 1.2

2 For the above variable X, what is the variance $\text{Var}(X)$?

a. 0.44

b. 0.50

c. 0.64

d. 0.84

3 If X is a binomial random variable with parameters $n=100$ and $p=0.2$, what is the standard deviation σ?

a. 4

b. 16

c. 20

d. 2

Section 2: Continuous Distributions & Probabilities

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4 A continuous random variable X is uniformly distributed over the interval $[2, 10]$. What is $P(4 \leq X \leq 7)$?

a. $\frac{3}{8}$

b. $\frac{1}{2}$

c. $\frac{5}{8}$

d. $\frac{1}{4}$

5 What is the expected value of a uniform random variable on $[a, b]$?

a. $\frac{a+b}{2}$

b. $\frac{b-a}{2}$

c. $\frac{a+b}{12}$

d. ab

6 In any continuous probability distribution, what is $P(X = c)$ for any single exact point c?

a. 0

b. 1

c. $\frac{1}{2}$

d. Undefined

Section 3: Conditional Probability & Joint Events

7 If $P(A) = 0.5$ and $P(B) = 0.4$, and A and B are mutually exclusive, what is $P(A \cap B)$?

a. 0

b. 0.20

c. 0.90

d. 0.10

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8 Given $P(A) = 0.6, P(B) = 0.3,$ and $P(A \cup B) = 0.7$, are events A and B independent?

a. No, because $P(A \cap B) = 0.20 \neq 0.18$

b. Yes, because $P(A \cap B) = 0.18$

c. Yes, because they are mutually exclusive

d. Cannot be determined

9 What is Bayes' formula for $P(A \mid B)$?

a. $\frac{P(B \mid A)P(A)}{P(B)}$

b. $\frac{P(A \mid B)P(B)}{P(A)}$

c. $\frac{P(A)P(B)}{P(A \cap B)}$

d. $P(A) + P(B)$

Section 4: Combinatorial Probability & Geometric Probability

10 A point is chosen at random inside a square of side 10. What is the probability that it falls inside the inscribed circle?

a. $\frac{\pi}{4}$

b. $\frac{\pi}{2}$

c. $\frac{\pi}{8}$

d. $\frac{1}{2}$

11 From a standard deck of 52 cards, what is the probability of being dealt a 5-card hand containing all 4 aces?

a. $\frac{48}{\binom{52}{5}}$

b. $\frac{4}{\binom{52}{5}}$

c. $\frac{1}{\binom{52}{5}}$

d. $\frac{48}{52}$

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12 Two real numbers x and y are chosen independently from $[0, 1]$. What is the probability that $x + y \leq 1$?

a. $\frac{1}{2}$

b. $\frac{1}{4}$

c. $\frac{3}{4}$

d. $\frac{1}{3}$

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